

Help, proving is hard!

1. Prerequisites

If you feel like you have trouble:

- checking whether a given proof has a fallacy, and
- producing your own proofs

despite having taken the Discrete Mathematics course, these notes are for you! Of course, the tips from `0-programming.pdf` also apply to proving:

- a mathematical claim is an interface. Its possible proofs are its implementations;
- claim-driven proving: understand the claim first, and only prove it later;
- hole programming: leave a hole if you are stuck and move to a different part.

2. Notation

For brevity, we can use symbols to refer to logical connectives and quantifiers. However, the problem statements can mix and match symbolic and keywords.

2.1. Connectives

All binary operators are right associative. From highest to lower priority:

$\wedge$	and
$\vee$	or
$\rightarrow$	implies
$\perp$	absurd

For each proposition P , we define $\neg P$ ("not P ") as:

$$\neg P := P \rightarrow \perp.$$

For each proposition P and Q , we define $P \leftrightarrow Q$ " P if and only if Q " as:

$$P \leftrightarrow Q := P \rightarrow Q \wedge Q \rightarrow P.$$

2.2. Quantifiers

$\forall$	for all
$\exists$	there exists

3. Premises vs goal

When reading a mathematical claim we are asked to prove, we can distinguish between assumptions and goal.

Assumptions (or premises) are what we already know, and can hence freely use in our proof. We can have zero, one, or more of them.

The **goal** is what we need to convince the reader of.

4. Mathematical claims as challenges

Consider an exercise asking you to prove a mathematical claim φ from a set Γ of mathematical facts. We can see this as a proof **challenge**: the professor is claiming that there exists some proof of φ using only assumptions in Γ , and they are asking you to find an example of such a proof. Much like $\Gamma' \vdash \varphi'$ means “there exists a derivation tree over premises set Γ' with root in φ' ” in $\mathcal{A}$, we could write down the professor’s claim as $\Gamma \vdash \varphi$. The only difference is that, in $\mathcal{A}$, we are looking for a derivation tree, while today we’ll be looking for a mathematical proof.

Definition 4.1: There are two kinds of *challenges*:

- $\Gamma \vdash \varphi$, which is an **open** challenge you need to solve;
- $\checkmark$, which is a **closed** challenge you just solved (by providing a correct proof).

Usually, we give a short name $H, K, L, M, N, \dots$ to each element of Γ .

5. Proofs as solutions

You read the problem statement carefully, and were able to tell apart the set of premises Γ from the goal φ . We have also given a name to each element of Γ for convenience, so that we write Γ as $H_1 : \varphi_1, \dots, H_n : \varphi_n$, usually leaving the parenthesis implicit. What is a proof for the open challenge $c_0 := \Gamma \vdash \varphi$?

Definition 5.1: A *proof* for a challenge c_0 is a finite list $s_0, c_1, s_1, c_2, s_2, \dots, c_{n-1}, s_{n-1}, c_n$ in which:

- each c_i is a challenge, with $c_n = \checkmark$;
- each s_i is a statement which allows us to move from c_0 to c_{i+1} , according to the rules below.

Here is an example (even though we haven’t learned about the rules mentioned above yet).

Example 5.1: If A and B , then B and A . $\vdash A \wedge B \rightarrow B \wedge A$

Proof: We assume $A \wedge B$ (H) and prove $B \wedge A$.

$H : A \wedge B \vdash B \wedge A$

We prove $B \wedge A$:

- $B: H : A \wedge B \vdash B$

To prove B , we show conjunction $A \wedge B$.

$H : A \wedge B \vdash A \wedge B$

True by H.

✓

- $A: H : A \wedge B \vdash A$.

To prove A , we show conjunction $A \wedge B$.

$H : A \wedge B \vdash A \wedge B$

True by H.



You can write the challenges (in blue) using a pencil, and only on the right side of the page, and delete them after the proof is complete. Note that the last proof statement features two nested proofs! Much like any other proof, these can be of arbitrary finite length.

6. Proof statements as moves

Each non-atomic formula has a **main symbols** (at the root of the syntax tree), whether it is a logical connective or a quantifier. Rules fall each main symbol fall into one of two categories:

- **introduction** rules, telling us how to prove a formula with such a main symbol when it is our goal;
- **elimination** rules, telling us when we can prove our goal by proving a formula with such a main symbol instead.

6.1. Basics

$\Gamma, H : \varphi \vdash \varphi$

True by H.

✓

Any previously proved result can be used as assumption (use the result's name, rather than H,K,...). Use "... φ , i.e. ψ ... (by ...)" to (un)fold definitions and if-and-only-ifs in premises or goals.

6.2. $\wedge$ ("and")

$\Gamma \vdash A \wedge B$

We prove $A \wedge B$:

- A : (prove $\Gamma \vdash A$ here);
- B : (prove $\Gamma \vdash B$ here).

✓

$\Gamma \vdash A$

To prove A , we show conjunction $A \wedge B$.

$\Gamma \vdash A \wedge B$

$\Gamma \vdash B$

To prove B , we show conjunction $A \wedge B$.

$\Gamma \vdash A \wedge B$

6.3. $\vee$ ("or")

$\Gamma \vdash A \vee B$

We show left disjunct A .

$\Gamma \vdash A$

$\Gamma \vdash A \vee B$

We show right disjunct B .

$\Gamma \vdash B$

$\Gamma \vdash \varphi$

We prove disjunction $A \vee B$, and proceed by cases on it:

- disjunction proof: (prove $\Gamma \vdash A \vee B$ here);
- case A : (prove $\Gamma, K : A \vdash \varphi$ here);
- case B : (prove $\Gamma, L : B \vdash \varphi$ here).

✓

6.4. $\rightarrow$ ("implies")

$\Gamma \vdash A \rightarrow B$

We assume A (H) and show B .

$\Gamma, H : A \vdash B$

$\Gamma \vdash \varphi$

To prove φ , showing A suffices:

- $A \rightarrow \varphi$: (prove $\Gamma \vdash A \rightarrow \varphi$);
- A holds: (prove $\Gamma \vdash A$).

✓

6.5. $\perp$ ("absurd")

$\Gamma \vdash \varphi$

We'll show this is absurd.

$\Gamma \vdash \perp$

6.6. $\forall$ ("for all")

$\Gamma \vdash \forall x \in X. P(x)$

Let $a \in X$ (H)¹. We show $P(a)$.

$\Gamma, H : a \in X \vdash P(a)$

$\Gamma \vdash P(a)$

To prove $P(a)$, we show $a \in X$ and $\forall x \in X. P(x)$:

- $a \in X$: (prove $\Gamma \vdash a \in X$);
- $\forall x \in X. P(x)$: (prove $\Gamma \vdash \forall x \in X. P(x)$).

✓

6.7. $\exists$ ("there exists")

$\Gamma \vdash \exists x \in X. P(x)$

We choose $x := t$, and show it satisfies the desiderata.

- $t \in X$: (prove $\Gamma \vdash t \in X$);
- $P(t)$: (prove $\Gamma \vdash P(t)$).

✓

$\Gamma \vdash a \in X \wedge P(a)$ ¹

We allowed ourselves to consider some $a \in X$ such that $P(a)$ due to the following proof of $\exists x \in X. P(x)$.

$\Gamma \vdash \exists x \in X. P(x)$

¹The variable a must be generic, so it cannot occur free in any assumption of Γ . It does not need to be the same as the one used by the quantifier in question.